

Vishesh Saluja

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Education

George Mason University , MS in Computer Science <i>Fairfax, VA, USA</i>	May 2025
Guru Gobind Singh Indrapratha University , B.Tech in Information Technology <i>New Delhi, India</i>	Jul 2023

Experience

Research Assistant – Generative Deep Learning , <i>George Mason University</i>	May 2025 – Present
- Working on molecular graph generation using MolGAN to design novel, valid molecules. - Integrating Birdie-style pretraining objectives to enhance diversity and synthesizability. - Aiming to generate high-quality molecular candidates suitable for lab-scale development and drug discovery pipelines.	
Software Developer & Research Assistant , <i>Geoweaver – NASA/ESIP</i>	Feb – Oct 2024
- Led 4 interns on ML workflows for geospatial data and co-authored their first publications. - Added support for PostgreSQL, MySQL, and H2 with CLI integration. Implemented an in-app code editor and improved UI responsiveness by 30%. - Reduced processing time by 25% in Snow/Ozone Estimation models and Ran large-scale workflows on Hopper supercomputer with 20% speed-up. - Introduced Vitest for unit testing, significantly accelerating development feedback loops and reducing CI runtime.	
Data Science Intern , <i>QBurst Technologies</i>	Jul – Aug 2022
- Improved BigQuery query performance by 15%. Conducted training and documentation improving team output by 10%. - Assisted in developing a pipeline to migrate companies Database from Amazon RedShift to BigQuery.	
ML Intern , <i>DynPro</i>	Apr – Jun 2022
- Optimized recommendation systems, leading to a 6.5% gain in prediction accuracy. - Delivered regression and clustering results boosting success rate by 20%.	
Software Engineer Intern , <i>Accops Systems</i>	Jan – Mar 2022
Worked on backend logic using Python and C++ and Reduced bug resolution time by 15%.	
Cloud Trainee , <i>Tech Explica</i>	Jul – Aug 2021
- Deployed multiple websites on AWS using EC2 and S3. - Increased deployment efficiency by 20% by implementing CI/CD pipelines with GitHub Actions.	

Publication

- Socioeconomic and Industrial Factors Influencing U.S. Ozone Pollution – NASA ESIP (AGU).
- Snow Water Equivalent Estimation Using ML & Remote Sensing – NASA ESIP.

Projects

Mason Critique – <i>Awarded 2nd Place</i>	HackFax 2025
- Developed a full-stack web app using React(Node.js, TypeScript), Firebase, and Tailwind CSS, enabling 50+ GMU students to review courses and professors with real-time feedback. - Engineered backend with Firestore & Google OAuth, integrating GPA calculators, grade charts (Chart.js).	
MindMate – <i>AI Mental Wellness Companion</i>	Feb – Mar 2024
- Built a full-stack mental health app in 36 hours using React, FastAPI, and Azure OpenAI (GPT-4o) to deliver empathetic chat, journaling, and mood-aware prompts while deploying a stable MVP with GPT-driven follow-up logic. - Integrated RAG-based response flow using vector search over emotional memory logs, enabling context-aware, personalized replies grounded in user history. - Designed with Framer Motion UI and a SQLite backend, achieving 95% response accuracy and storing 100+ emotional logs securely.	
Diabetic Retinopathy Detection – <i>ML on Retinal Images</i>	Apr – May 2023
Used EfficientNet-B4, ResNet18, and custom CNN with QWK: 0.69.	
Brain Hemorrhage Detection – <i>Binary ResNet Classifier</i>	Dec 2022
Built ResNet18 to classify hemorrhagic vs normal CT scans.	

Skills and Tools

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| <ul style="list-style-type: none">Languages: Python, C, C++, SQL, Java, BashLibraries/Tools: PyTorch, TensorFlow, Pandas, NumPy, Docker, Kubernetes, GitHub Actions, Terraform | <ul style="list-style-type: none">Web & Cloud: HTML, CSS, JS, React, Angular, jQuery, AWS, Azure, GCP, ML/DL, Security |
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